Prune Strategy to YOLO-Based Model

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***Abstract*—One-stage networks have been widely used in target detection systems that need to learn from the big data. Most of them have good real-time performance and accuracy. However, due to their convolution structure, in actual applications, they still require more memory space and stronger computing power, thereby we choose to apply a pruning strategy on the target detection network to reduce the size of the network model and the number of parameters. In order to prove the practicability of the pruning method, we select the YOLOv3 model for experiments and provide a crop data set collected in the orchard to show the model effect. In this data set, in the best case, the volume of the network model is reduced by 52.7% compared with the original model, and the reasoning time is reduced by 57.5%. At the same time, it also uses data processing methods to compensate for the decrease in accuracy caused by pruning.**

Keywords—prune, crop data set, YOLOv3-based model, reasoning time

I. Introduction

The majority of CNN-based object detectors are widely used in recommendation systems. For example, searching for the actual situation of fruit trees via orchard camera, whereas the quantity of fruits in a unit can be gained through one target detection system. The object detectors in real world applications are mostly constrained by model size, run-time memory and number of computing operations, but actually, only parts of them satisfy the above three demands. The state-of-the-art object detectors, including single stage SSD [1], YOLO [2] or multiple stage R-CNN [3] series like R-FCN [4], and many variations or extended work [5]. For those detectors running on CPU or GPU platform, there backbone could be VGGNet [6], DenseNet [7], CSPNet [8] or MobileNet [9,10], or ShuffleNet [11]. Comparison of the YOLOv3 and other state-of-the-art object detectors, the former operate in real-time and requires less computation amounts. Therefore, we use the YOLOv3-based models in our experiments.

The main goal of this work is finding the YOLOv3-based model, which meets both real-time and exact, and for which methods that only change the training strategy or increase the training cost can improve network's accuracy. Our contributions are summarized as follows:

1) We furnish a complex data set for model's training. The data set is collected by the Unmanned Aerial Vehicle (UAV) at different distances, including three weather conditions: sunny, cloudy, and foggy, and contents the concepts of dense data set.

2) We contrast the effect of different YOLOv3-based models, including the time taken to detect a single picture, the mAP value and other important indicators on the entire test set.

3) We verify influence of data preprocessing methods during the detector training, and judged which method can improve the mAP value obviously.

The rest of this paper is organized as follows. First, we briefly introduce the characteristics of YOLOv3-based models. Second, the proposed models are detailed. Third, the provided data set and the experimental results are benchmarked. Finally, we will summarize this work.

II. The Proposed Model

In this section, we present the YOLOv3-based models' improvements, and then recommend the implementation process of YOLOv3-prune.

A. YOLO-based Models

As a one-stage series network, YOLOv3 has a higher accuracy and faster processing speed, but in actual scenarios, it requires large number of GPUs for training with a large mini-batch-size. Therefore, some network structures based on YOLOv3 have appeared.

First, the traditional YOLOv3 model optimized with this method include three different scales to detect, increasing the detection effects on small objects, modifying the loss function, predicting the object confidence and category through logistic regression, and employing the multi-label classification instead of SoftMax classification. The complete network structure is shown in the Figure1. Second, YOLOv3-tiny is designed to enhance the detection speed. The imparity with YOLOv3 is that: it reduces the amount of convolutional structures, and fits out only two detection layers, which significantly shortens inference time and brings an evident drop in accuracy.

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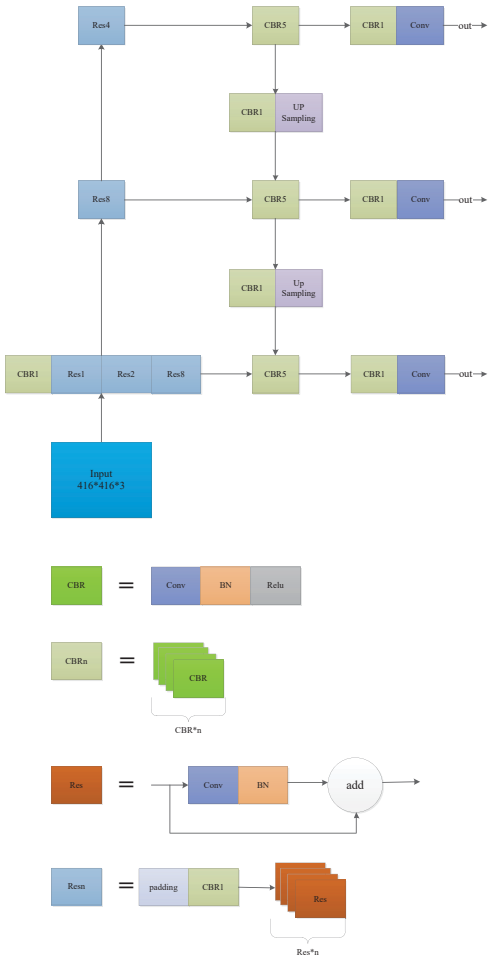


Fig. 1. This diagram clearly introduces the network structure of YOLOv3, and we can find that the BN layer, the convolution layer and the activation layer together constitute a component CBR of YOLOv3

B. Pruning Process

Pruning refers to the removal of channels that have less impact on the network after the sparse training, thereby reducing the amount of network parameters. On account of applying YOLOv3 to actual scenes, we should prune on the basis of retaining accuracy constant.

In the design of YOLOv3, each convolution structure is composed of three network layers: convolution layer, BN layer, and activation layer, which is showed in Figure 1, and they are called CBR. The role of the BN layer is to normalize the previous convolutional layer's output, and this normalization runs through the entire neural network. On the one hand, this normalization forcibly alters the input into a relatively normal distribution with mean 0 and variance 1 so that the output from BN layer can be used easily for the next Relu layer to avoid gradient disappearance. On the other hand, the fourth putting of BN associates all the samples in a mini-batch together, thus the network no longer producing deterministic values for a given training example [12,13]. It is precisely because of the BN layer's characters that runs through the entire network and only changes the distribution, after pruning BN layer, the overall learning process may not be affected greatly. Therefore, we choose to prune YOLOv3's BN layer.

In this article, YOLOv3-prune will be divided into the following two steps:

1) Conduct sparse training of YOLOv3. In YOLOv3, assuming C_{out} and B_{out} respectively represent the previous convolutional layer's output and the BN layer's output, b_{mini} denotes the current mini-batch, and BN layer undergo the following transformation:

$$\hat{B} = \frac{C_{out} - \beta_{b_{mini}}}{\sqrt{\alpha^2_{b_{mini}} + t}}; B_{out} = \varepsilon * \hat{B} + \theta \quad (1)$$

Among them, $\beta_{b_{mini}}$ and $\alpha_{b_{mini}}$ are the mean and variance deviation values of input activations over b_{mini}, ε and θ are two parameters that can be learned. This conversion is performed at the BN layer, providing the possibility of linearly transforming normalized activations back to any scales. Owing to ε is a parameter that scales $\hat{B}$, we directly treat ε as the object of network pruning. In the first step, we ought to conduct sparse training by adding L1-norm to B_{out} as the constraint condition for its forward propagation during the update process of BN layer[14], and calculate the value of each BN layer's ε parameter as the basis for judging the influence of this channel on the network. In addition, the absolute value function added by L1 regularization creates the loss function cannot differentiable everywhere, and then the weight matrix can be generated into a sparse matrix through the back-propagation algorithm.

2) Pruning the weight files after training. For the created weight files, according to the sequence number of BN layer to find its corresponding weights firstly, this weight saves the ε value judged whether this channel is crucial. Secondly, the value of weights will be sorted from smallest to largest, and then, on the basis of pruning ratio γ to pruned. For example, the pruning ratio γ is 0.7, then put the top 70% of the weight value into 0, only join a bias θ. When all the weight information of BN layers is estimated, update the weight file through the formed arguments as our trained model, and generate a new network configuration file

In fact, there are two pruning methods: normal pruning and regular pruning. Normal pruning means strictly set the ε parameter to 0 according to γ, and obtain a considerable compression ratio. Regular pruning is designed for hardware deployment, and the number of filters after pruning is a multiple of 8. Regular pruning sacrifices part of the compression ratio for application, but at the same compression rate, the mAP value of regular pruning is generally higher than that of normal pruning. And we can clearly see the above pruning process in Figure 2.

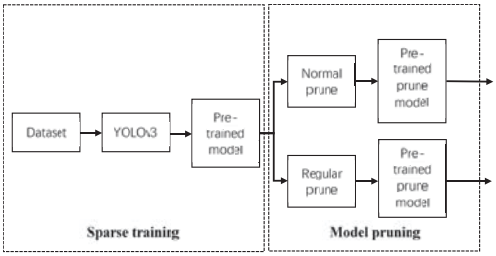


Fig. 2. This diagram describes the process from network sparsity training to pre-training model generation and pruning model generation through two different pruning methods.

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III. EXPERIMENTS

In order to get the most suitable pruning method and data preprocessing method in this data set, we will test the different YOLOv3-based models, and find which data preprocessing method can have a better effect.

A. Description of Data Set

The samples of the data set used in this article comes from a thousand of acres of orchards. The main crops are oranges, which are obtained by drone aerial photography. The size of the collected pictures is all 1920 * 1080, which retains the information of the object to the greatest extent.

The data set has the following three characteristics:

1) The data collection selects the three local weather conditions that are more common in the local area, which are sunny, cloudy and foggy. Among them, the data captured on sunny days has 285 pictures, and the data captured on cloudy days has 248 pictures, and here are 241 pictures taken by foggy, which ensure that the total number of data under the three environments is nearly equal.

2) A total of 778 effective pictures in this data set, including over 70000 oranges, meet the standards of dense data sets, and also meet the actual scene of the orchard when the crops mature.

3) Due to the intensive crops, during the data collection process, the drone shoots at three different angles: flat angle, depression angle, and elevation angle, which retains the spatial information of the target to a certain extent and can solve the target's occlusion problem to a certain extent.

B. Performance of Different Models

We will first test three YOLOv3-based models in the orange dataset. The evaluation criteria of the test are to consider both real-time and accuracy, so we compare the performance of the network models generated from the four perspectives of network parameter volume, model volume, time spent inferring a single picture, and mAP. We will train on a GTX1660Ti GPU and a i7 9750H CPU and always use a fixed (416 * 416)-pixel size as input data for training, and do not use any other data preprocessing methods, in addition, the YOLOv3-prune uses normal pruning, and the pruning rate is 0.3 for training and testing.

As for the performance of the network model generated by different pruning methods and different pruning rates on the orange data set, we also conducted a detailed test. The performance of YOLOv3, normal prune YOLOv3 and regular prune YOLOv3 on the test set is shown in Table I and Table II, which details the performance of the two pruning methods and the network model generated by different pruning rates on the orange dataset. In the comparative experiment, we compared all the pruned networks with the unpruned YOLOv3 network. By comparing the experimental results, we can find that, under the same pruning rate, the model volume of the normal pruning to the network model is about 8% smaller than that of the model after the regular pruning to the network model. At the same time, the normal pruning model's inference time on a single picture is about 25% shorter than that of the model after regular pruning. In addition, we can find that no matter whether normal pruning or regular pruning is used, the mAP value of the network model generated by them on the orange data set has dropped to a certain extent.

TABLE I. COMPARE THE PERFORMANCE OF DIFFERENT MODELS

Model	**Model processing method**			**Model evaluation**	
	Normal prune	***Regular prune***	***pruning radio***	***Model volume***	***Compr--ession ratio***
YOLO@416				492.6MB	
YOLO@416	√		0.3	337.9MB	68.6%
YOLO@416	√		0.5	233.0MB	47.3%
YOLO@416		√	0.3	375.4MB	76.2%
YOLO@416		√	0.5	268.0MB	54.4%

TABLE II. COMPARE THE PERFORMANCE OF DIFFERENT MODELS

Model	**Model processing method**			**Model evaluation**	
	Normal prune	***Regular prune***	***pruning radio***	***Time consuming***	***mAP***
YOLO@416				0.040	75.27
YOLO@416	√		0.3	0.018	73.37
YOLO@416	√		0.5	0.017	72.94
YOLO@416		√	0.3	0.023	73.88
YOLO@416		√	0.5	0.021	74.68
Faster RCNN@416				0.065	77.27
Mask RCNN@416				0.073	77.15

In comparison, when the pruning rate γ is 0.3, the mAP value of the model using regular pruning is reduced by 0.7% than the mAP value of the model using normal pruning; when the pruning rate γ is 0.5, the mAP value of the model using regular pruning reduces the mAP value of the model using normal pruning by 2.4%. On the other hand, for the same pruning method, the more intuitive comparison result is that as the pruning rate increases, the volume of the model generated after pruning will become smaller, the inference time will decrease, and the mAP will also decrease.

By comparison, we found that when the input image size is 416*416, the normal pruning method is used, and the model file generated with a pruning rate γ of 0.3 not only has a compression rate of 31.4%, but the time for inferring a

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single picture also decreases after 55%, mAP also has only dropped by 2.5%, which has little effect on the actual application of target detection, and will not affect the detection effect of the network too much.

Therefore, we choose the model file under this pruning condition to add data preprocessing methods to continue the test.

C. Data Preprocessing

YOLOv3 is a one-stage neural network, and data preprocessing plays a vital role in network training. Therefore, we will determine which method is suitable for this data by testing the impact of the following five data preprocessing methods on YOLOv3-prune.

1) Random shape training: The sample pictures of this training set are all taken from the scene of the drone, and the sizes are all 1080 * 1920. When using this method, we will adjust the size of the input batch size number of images to the H set A size in, where the H set is {320,352,384,416,448,480,512,544,576,608}.

2) Random angel training: When using this method, we will set the angel value to 5, and at this time, before the training picture is input to the network, it will be flipped within the range of -5 ° to +5 °.

3) Change saturation training: When using this method, we will set the saturation value to 1.5. At this time, before the training picture is input to the network, the saturation will be adjusted to 1.5 times the original picture.

4) Change exposure training: When using this method, we will set the exposure value to 1.5, and then adjust the exposure to 1.5 times the original image before the training image is input to the network.

5) Change hue training: When using this method, we will set the hue value to 0.1. At this time, before the training picture is input to the network; the chrome will be adjusted to 0.1 times the original picture.

In addition to random graphics training, we will not change the size of the input training picture. For the other four methods, we will take 416*416 size picture as input, and take normal pruning on the 0.3 pruning ratio for the train and test. After applying these five methods in the data set and using the designed network for training, the test results are shown in Table III.

TABLE III. COMPARISON OF DIFFERENT PRETREATMENT METHODS

Approach	Model evaluation		
	mAP	*Δ*	*Cumu Δ*
baseline	74.68	0	0
+ Random shape training	75.00	+0.32	+0.32
+ Random angel training	74.79	+0.11	+0.42
+ Change saturation training	74.71	+0.03	+0.44
+ Change exposure training	74.75	+0.07	+0.50
+ Change hue training	74.74	+0.06	+0.54

By comparison, we can find that after adding Random shape training, the accuracy of the network is the most obvious. When the three methods of Change saturation training, Change exposure training and Change hue training are used at the same time, the improving of accuracy is not simply to use them in their respective superimpose the effect of improving the network accuracy, indicating that the role of these three methods in this data set and the network structure has a certain repeatability. And Random angel training with Random shape training, although used together, also there will be a certain decrease in the accuracy of the increase due to the repetition of the effect, but the decrease is extremely small and can be ignored. Therefore, in order to reduce the training time of the network on the basis of ensuring the improvement of accuracy, and keep the original features of the data set as much as possible, we only choose the three data preprocessing methods of Random shape training, Random angel training and Change saturation training together.

IV. CONCLUSION

In this paper, we provide a crop data set collected in actual scenarios, and compare the model volume, inference time, and parameter amount on the test set of the model files generated after training on this data set by comparing several YOLOv3 structures that are more widely used. As well as the mAP value, YOLOv3-prune is the most suitable model for this data set. By comparing the performance of model files generated by two different pruning methods and different pruning rates on the test set, we choose the most appropriate pruning method and pruning rate. After that, different data preprocessing methods are added to the data set, and the selected pruning method and rate are used for training to make a model file. Through testing, three data preprocessing methods are selected and combined to help the network performance most. All existing data and training results can be reproduced.

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